

solution 2 (resubmission)

Apoorv Shah

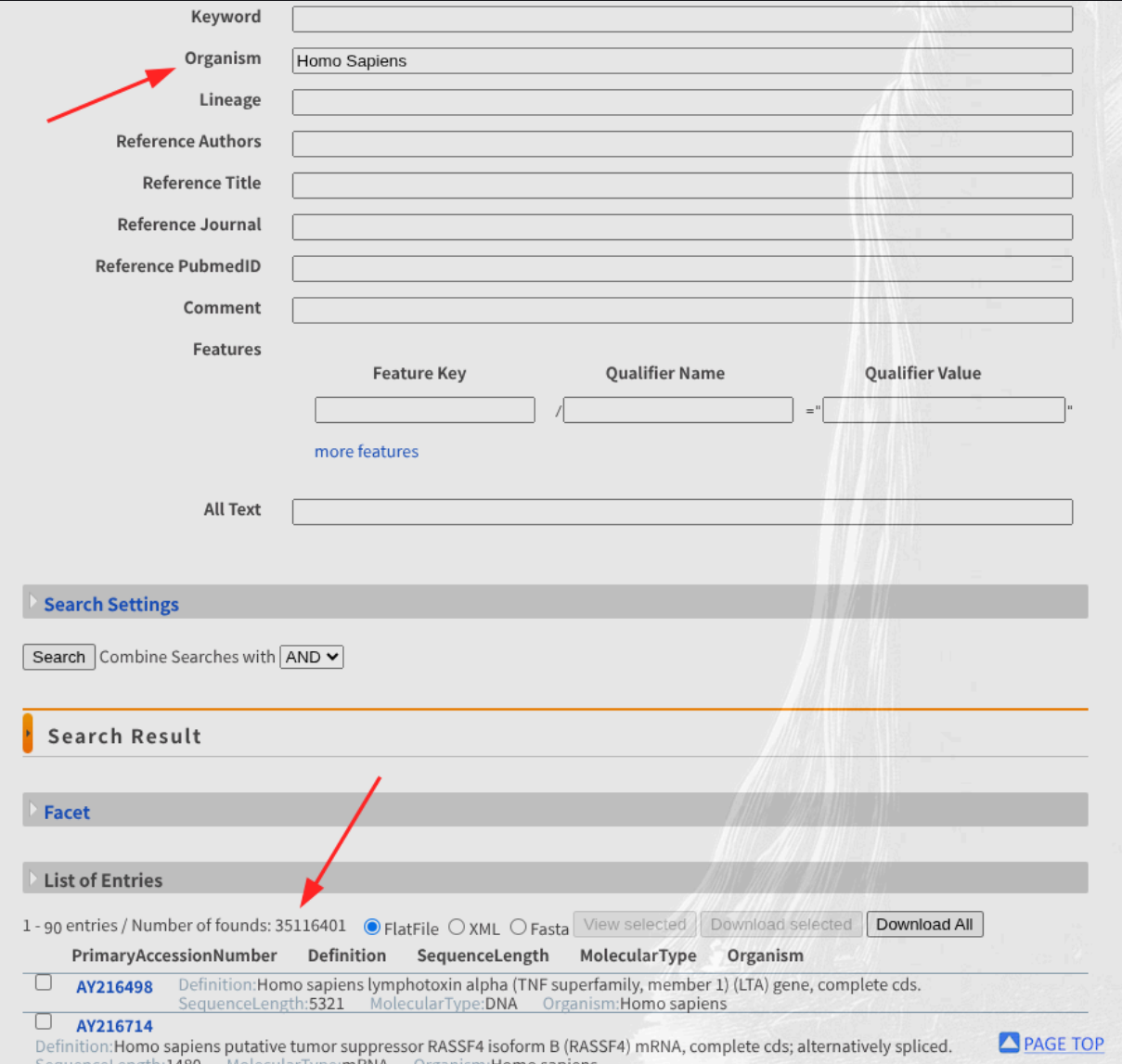
BS24B006

(sorry for sharing assignment, and extremely grateful for getting to resubmit)

Q1

DDBJ

this link opens the DDBJ advanced search for *Homo sapiens*



Keyword

Organism

Lineage

Reference Authors

Reference Title

Reference Journal

Reference PubmedID

Comment

Features

Feature Key	Qualifier Name	Qualifier Value

[more features](#)

All Text

Search Settings

Combine Searches with

Search Result

Facet

List of Entries

1 - 90 entries / Number of founds: 35116401 ☒ FlatFile ☐ XML ☐ Fasta

PrimaryAccessionNumber	Definition	SequenceLength	MolecularType	Organism
☐ AY216498	Definition:Homo sapiens lymphotoxin alpha (TNF superfamily, member 1) (LTA) gene, complete cds. SequenceLength:5321 MolecularType:DNA Organism:Homo sapiens			
☐ AY216714	Definition:Homo sapiens putative tumor suppressor RASSF4 isoform B (RASSF4) mRNA, complete cds; alternatively spliced. SequenceLength:1480 MolecularType:mRNA Organism:Homo sapiens			

[PAGE TOP](#)

so DDBJ = 35,116,401

GenBank



Nucleotide

Nucleotide

Homo Sapiens[Organism]



Search

Create alert

Advanced

Help

Species

Animals (29,784,649)
Plants (1)
Fungi (1)
Viruses (4,548)
Customize ...

Summary 20 per page Sort by Default order

Send to:

Filters: [Manage Filters](#)

Results by taxon

Top Organisms [\[Tree\]](#)

Homo sapiens (29781320)
synthetic construct (57325)
Mus musculus (3176)
Human papillomavirus 16 (78)
Trypanosoma cruzi (21)
All other taxa (78)
[More...](#)

Find related data

Database:

Select

[Find items](#)

Search details

Items: 1 to 20 of 29841998

<< First < Prev Page 1 of 1492100 Next > Last >>

- ☐ [Homo sapiens N-sulfolglucosamine sulfohydrolase \(SGSH\), RefSeqGene on chromosome 17](#)

26,661 bp linear DNA

Accession: NG_008229.2 GI: 2310189229

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

- ☐ [Homo sapiens FASTK mitochondrial RNA regulator 3 \(FASTKD3\), transcript variant 3, non-coding RNA](#)

1,001 bp linear transcribed-RNA

Accession: NR_073608.2 GI: 1890392847

[PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

Molecule types

genomic
DNA/RNA (16,937,41)
mRNA (9,950,426)
rRNA (788)
Customize ...

Source databases

INSDC (GenBank)
(29,256,885)
RefSeq (424,468)
Customize ...

Sequence Type

in the nucleotide database in genbank it shows 29,841,998

EMBL

this link

ebi.ac.uk/ena/browser/advanced-search

Relaunch to update

Data Type Query Inclusion/Exclusion Fields Data Filters Results

Download ENA records: FASTA Download report: JSON

TEXT TSV

Accession	Description
A00383	H.sapiens NANB-hepatitis associated DNA
A06851	H.sapiens gene for relaxin, exon 1
A08002	Nucleotide sequence 7 from patent number EP0370719
A08905	H.sapiens (haplotype 2B, allele MS32, isolate English, serial number 31) minisatellite sequence
A11954	H.sapiens mRNA fragment for calcitonin
A14571	(2'-5') oligo A synthetase cDNA for 1.6kb RNA (E16,clone 9-21)
A25971	H.sapiens mRNA for T-cell receptor IGR b 20 Vbeta 9
A26249	Human shIL5R-alpha gene
A26437	Human granzyme B
A28104	Human GABAa receptor alpha-5 subunit

Items per page: 10 1 - 10 of 10000 (29244316 total)

New Search Back Copy Curl Request Copy Http Request

so 29,244,316

finally:

- ◆ DDBJ = 35,116,401
- ◆ GenBank = 29,841,998
- ◆ EMBL = 29,244,316

[New comment](#)

even though all three sync under INSDC every single day, the counts don't match up 1:1 mainly because of how each node indexes raw ESTs/bulk reads, updates their search engines, and filters patent or patent-related sequences differently

Q.2

link

this is the sequence taken from genbank:

Plaintext

```
ATGAGGAAGAAGCGGAGACAGCGACGAAGAATGAAGAAGGTCTTTGAAA
GGTGTGAGTTGGCCAGAACTC
TGAAAAGATTGGGAATGGATGGCTACAGGGGAATCAGCCTAGCAAACCTG
GATGTGTTTGGCCAAATGGGA
GAGTGGTTACAACACACGAGCTACAACTACAATGCTGGAGACAGAAGC
ACTGATTATGGGATATTTTCAG
ATCAATAGCCGCTACTGGTGTAAATGATGGCAAAACCCCAGGAGCAGTTA
ATGCCTGTCATTTATCCTGCA
GTGCTTTGCTGCAAGATAACATCGCTGATGCTGTAGCTTGTGCAAAGAG
GGTTGTCCGTGATCCACAAGG
CATTAGAGCATGGGTGGCATGGAGAAATCGTTGTCAAACAGAGGTGTC
CGTCAGTATGTTCAAGGTTGT
GGAGTGTAATGA
```

using emboss we can figure out the GC content

The screenshot shows the GEECEE web interface. The left sidebar contains a menu with categories like ALIGNMENT, DATA RESOURCES, and UTILS. The main area is titled 'GEECEE' and 'Calculate fractional GC content of nucleic acid sequences'. It has an 'input section' with a text area containing the sequence 'AY330867 ATGAGGAAGAAGCGGAGACAGCGACGAAGAATGAAGAAGGTCTTTGAAAGGTGTGAGT'. Below this is an 'output section' with a text box for 'outfile file name'. The 'Execution mode' is set to 'interactive'. At the bottom, a status bar shows 'ay330867.geecee cmd' and a table with the following data:

#Sequence	GC content
AY330867	0.47

so it is 47%

the total length of the sequence is 432 bp, and with G+C adding up to 203 bases, $(203 / 432)$ gives right around 0.4699, so 47% is correct

Q.3

- ◆ <https://www.ncbi.nlm.nih.gov/nuccore/33088589>
- ◆ https://getentry.ddbj.nig.ac.jp/getentry/na/AY330867/?format=flatfile& filetype=html&trace=true&show_suppressed=false&limit=10
- ◆ <https://www.ebi.ac.uk/ena/browser/view/AY330867?show=orcid-links>

all databases give the sequence, translation and other useful info

all three are part of the INSDC and sync daily, so the core biological sequences and accession numbers are identical across all of them

(taken from ddbj)

```
LOCUS       AY330867                432 bp    mRNA    linear    SYN 29-JUL-2004
DEFINITION   Synthetic construct human lysozyme mRNA, complete cds.
ACCESSION    AY330867
VERSION      AY330867.1
KEYWORDS     .
SOURCE       synthetic construct
  ORGANISM   synthetic construct
              other sequences; artificial sequences.
REFERENCE    1 (bases 1 to 432)
AUTHORS      Xie,X.D., Chen,Z.H. and Zhou,C.S.
TITLE        Direct Submission
JOURNAL      Submitted (26-JUN-2003) Postdoctoral Scientific Research Station of
              Gansu Yasheng Industrial (Group) Co., Ltd., Zhangye Road, Lanzhou,
              Gansu 730030, P.R.China
FEATURES             Location/Qualifiers
     source           1..432
                     /organism="synthetic construct"
                     /mol_type="mRNA"
                     /db_xref="taxon:32630"
     CDS              1..429
                     /codon_start=1
                     /transl_table=11
                     /product="human lysozyme"
                     /protein_id="AAP93336.1"
                     /translation="MRKKRRQRRRMKKVFERCELARTLKRLGMDGYRGISLANWMCLA
                     KWESGYNTRATNYNAGDRSTDYGFQINSRYWCNDGKTPGAVNACHLSCSALLQDNIA
                     DAVACAKRVVRDPQGIRAWVAWRNRCQNRGVRQYVQGCGV"
BASE COUNT        131 a                72 c                129 g                100 t
ORIGIN
1 atgaggaaga agcgagaca gcgacgaaga atgaagaagg tctttgaaag gtgtgagttg
61 gccagaactc tgaaaagatt gggaatggat ggctacaggg gaatcagcct agcaaaactg
121 atgtgttttg ccaaattgga gagtgggttac aacacacgag ctacaaacta caatgctgga
181 gacagaagca ctgattatgg gatatttcag atcaatagcc gctactgggt taatgatggc
241 aaaaccccag gacgagttaa tgcctgtcat ttatcctgca gtgctttgct gcaagataac
301 atcgtctatg ctgtagcttg tgcaaaagg gttgtccgtg atccacaagg cattagagca
361 tgggtggcat ggagaaatcg ttgtcaaaac agagggtgcc gtcagtatgt tcaaggttgt
421 ggagtgtaat ga
//
```

but they have some differences like:

- ◆ genbank and ddbj use full words like locus and origin for headers but embl uses two letter codes like id and sq
- ◆ ddbj clearly shows the exact base count right before the sequence but genbank skips it completely
- ◆ embl web view gives extra stuff like orcid data claims and cross references to other databases
- ◆ genbank and ddbj put the sequence numbers on the left side but embl puts them on the far right
- ◆ genbank it is tied to NCBI, meaning it natively cross-links to PubMed for literature, BLAST for alignments, and RefSeq for curated genomes
- ◆ embl it is built into the european framework, directly linking nucleotide sequences to Ensembl (genome browsers) and UniProt (protein databases)
- ◆ ddbj it serves the asian research node and links out to specific functional databases like KEGG for metabolic pathway analysis

the raw sequence data is exactly the same; the real difference is just the web interface and the downstream analytical tools each database connects us to

New comment

basically the actual FASTA/nucleotide string is identical line-for-line across all three, it's just the flatfile parser syntax and which bioinformatics ecosystem you get plugged into that changes between them

Q.4

papers link

discrimination of beta barrel membrane proteins

Scholar About 20,500 results (0.41 sec) YEAR

A Hidden Markov Model method, capable of predicting and discriminating β -barrel outer membrane proteins [PDF] springer.com
 PG Bagos, TD Liakopoulos, IC Spyropoulos... - BMC ..., 2004 - Springer
 ... **membrane proteins** are divided into two distinct structural classes, the α -helical **membrane proteins** and the β -barrel **membrane proteins**... **proteins** of that type are located mostly in the cell ...
 ☆ Save Cite Cited by 234 Related articles All 17 versions

[HTML] TMBETADISC-RBF: **discrimination of β -barrel membrane proteins** using [HTML] sciencedirect.com
 RBF networks and PSSM profiles
 YY Ou, MM Gromiha, SA Chen, M Suwa - Computational biology and ..., 2008 - Elsevier
 ... **discrimination** of β -barrel or outer **membrane proteins** (OMPs) from other folding types of globular and **membrane proteins** ... have been proposed for **discriminating** OMPs based on ...
 ☆ Save Cite Cited by 106 Related articles All 7 versions

Predicting transmembrane beta-barrels in proteomes [PDF] oup.com
 HR Bigelow, DS Petrey, J Liu, D Przybylski... - Nucleic acids ..., 2004 - academic.oup.com
 ... **beta-barrel membrane proteins** directly from sequence. One reason is that only very few high-resolution structures for **transmembrane beta-barrel (TMB) proteins** ... **protein discrimination** ...
 ☆ Save Cite Cited by 219 Related articles All 16 versions

PRED-TMBB2: improved topology prediction and detection of β -barrel outer membrane proteins [PDF] oup.com
 KD Tsirigos, A Elofsson, PG Bagos - Bioinformatics, 2016 - academic.oup.com
 ... The PRED-TMBB method is based on Hidden Markov Models and is capable of predicting the topology of **beta-barrel** outer **membrane proteins** and **discriminate** them from water-...
 ☆ Save Cite Cited by 103 Related articles All 12 versions

[PDF] A sequence-profile-based HMM for predicting and **discriminating beta barrel membrane proteins** [PDF] academia.edu
 PL Martelli, P Fariselli, A Krogh, R Casadio - ISMB, 2002 - academia.edu
 ... In this paper we use the prototypes of the β -barrel **membrane proteins** crystallized so far for training and testing a method based on HMM. The model is novel and is trained on ...
 ☆ Save Cite Cited by 249 Related articles All 13 versions

Functional discrimination of membrane proteins using machine learning techniques [PDF] springer.com
 MM Gromiha, Y Yabuki - BMC bioinformatics, 2008 - Springer
 ... of 64% in a data set of 1718 **membrane proteins**. The application of amino acid occurrence ... have **discriminated** transporters from other α -helical and β -barrel **membrane proteins** with ...
 ☆ Save Cite Cited by 93 Related articles All 16 versions

some examples:

- ◆ Liu, Q., Zhu, Y., Wang, B., & Li, Y. (2003). Identification of β -barrel membrane proteins based on amino acid composition properties and predicted secondary structure. *Computational Biology and Chemistry*, 27, 355–361.

[https://doi.org/10.1016/s1476-9271\(02\)00085-3](https://doi.org/10.1016/s1476-9271(02)00085-3) Cited by: 52

- ◆ Ou, Y.-Y., Gromiha, M. M., Chen, S.-A., & Suwa, M. (2008). TMBETADISC-RBF: Discrimination of β -barrel membrane proteins using RBF networks and PSSM profiles. *Computational Biology and Chemistry*, 32, 227–231.

<https://doi.org/10.1016/j.compbiolchem.2008.03.002> Cited by: 106

The screenshot shows the PubMed website interface. At the top, there's a header with the NIH logo and 'National Library of Medicine' text. Below that, the PubMed logo is visible. A search bar contains the query 'discrimination of beta barrel membrane proteins'. To the right of the search bar are buttons for 'Advanced', 'Create alert', 'Create RSS', and 'User Guide'. Below the search bar, there are buttons for 'Save', 'Email', and 'Send to'. To the right of these buttons, there's a 'Sort by' dropdown set to 'Best match' and a 'Display options' button. The main content area shows '54 results' and a pagination bar indicating 'Page 1 of 6'. On the left side, there are filters for 'MY CUSTOM FILTERS', 'RESULTS BY YEAR' (with a bar chart showing results from 1992 to 2026), 'PUBLICATION DATE' (with radio buttons for 1 year, 5 years, 10 years, and Custom Range), and 'TEXT AVAILABILITY' (with checkboxes for Abstract and Free full text). The search results list two articles found by citation matching. The first article is 'TMBETADISC-RBF: Discrimination of beta-barrel membrane proteins using RBF networks and PSSM profiles' by Ou YY, et al. (2008). The second article is 'Current developments on beta-barrel membrane proteins: sequence and structure analysis, discrimination and prediction' by Gromiha MM, et al. (2007). A yellow warning box states 'Quoted phrase not found in phrase index: "discrimination of beta barrel membrane proteins"'. At the bottom, there's a link to 'OMPdb: A Global Hub of Beta-Barrel Outer Membrane Proteins'.

- ◆ Tsirigos, K. D., Elofsson, A., & Bagos, P. G. (2016). PRED-TMBB2: improved topology prediction and detection of beta-barrel outer membrane proteins. *Bioinformatics*, 32, i665–i671. <https://doi.org/10.1093/bioinformatics/btw444> Cited by: 103

[New comment](#)

predicting beta-barrels is way harder than alpha-helical proteins because the hydrophobic pattern alternates every single residue (facing lipid vs lumen) rather than staying in continuous hydrophobic stretches

Q.5

Kihara D Google Scholar link

one paper published in nature journal example

paper link

they report the results from the first large-scale community-based critical assessment of protein function annotation (CAFA) experiment.

on pubmed

 An official website of the United States government [Here's how you know](#) ✓

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

PubMed Advanced Search Builder

PubMed
User Guide

Add terms to the query box

Author Kihara D. ×

ADD ▾

Show Index

Query box

Enter / edit your search query here

Search ▾

An official website of the United States government [Here's how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

Log in

PubMed

Kihara D.[Author] × **Search**

[Advanced](#) [Create alert](#) [Create RSS](#) [User Guide](#)

[Filters](#) [Timeline](#) Sort by: [Best match](#) [Display options](#)

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275 results << < Page **1** of 28 > >>

i It looks like you are searching for an author. Results are currently sorted by Best Match. To see the newest results first, [change the sort order to Most Recent](#).

☐ [Effect of phosphorylation barcodes on arrestin binding to a chemokine receptor.](#)

1 Chen Q, Schafer CT, Mukherjee S, Wang K, Gustavsson M, Fuller JR, Tepper K, Lamme TD, Aydin Y, Agrawal P, Terashi G, Yao XQ, **Kihara D**, Kossiakoff AA, Handel TM, Tesmer JJG.
Cite Nature. 2025 Jul;643(8070):280-287. doi: 10.1038/s41586-025-09024-9. Epub 2025 May 21.
PMID: 40399676 [Free PMC article](#).

☐ [Computational protein function predictions.](#)

2 **Kihara D.**
Cite Methods. 2016 Jan 15;93:1-2. doi: 10.1016/j.jymeth.2016.01.001.
PMID: 26778120 No abstract available.

☐ [Computational prediction of disordered binding regions.](#)

3 **Kihara D.**

[New comment](#)

CAFA is basically the CASP equivalent for automated protein function prediction, testing algorithms blindly across Gene Ontology categories (molecular function, biological process, cellular component)

Q.6

paper link

"Mitochondrial beta-barrel proteins, an exclusive club?"

there are lots of related articles



Cell 2008 Dec 26;135(7):1158-9?



Search

[Advanced](#) [Create alert](#) [Create RSS](#)

[User Guide](#)

Found 1 result for an alternative search.

Your search for *Cell 2008 Dec 26;135(7):1158-9?* retrie...

Save

Email

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Display options 

Comment

> Cell. 2008 Dec 26;135(7):1158-9; author reply 1159-60.

doi: 10.1016/j.cell.2008.12.017.

Mitochondrial beta-barrel proteins, an exclusive club?

Kenichiro Imai, M Michael Gromiha, Paul Horton

PMID: 19109882 DOI: 10.1016/j.cell.2008.12.017 

[Free article](#)

 Full text links

 Cite

No abstract available

[PubMed Disclaimer](#)

Comment on

Similar articles

[Transport proteins \(carriers\) of mitochondria.](#)

Wohlrab H.

IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.

PMID: 18816452 Review.

[Systematic analysis of the twin cx\(9\)c protein family.](#)

Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.

J Mol Biol. 2009 Oct 23;393(2):356-68. doi: 10.1016/j.jmb.2009.08.041. Epub 2009 Aug 21.

PMID: 19703468

[Mitochondrial permeability transition pore opening as a promising therapeutic target in cardiac diseases.](#)

Javadov S, Karmazyn M, Escobales N.

J Pharmacol Exp Ther. 2009 Sep;330(3):670-8. doi: 10.1124/jpet.109.153213. Epub 2009 Jun 9.

PMID: 19509316

[Vitamin E protects against the mitochondrial damage caused by cyclosporin A in LLC-PK1 cells.](#)

de Arriba G, de Hornedo JP, Rubio SR, Fernández MC, Martínez SB, Camarero MM, Cid TP.

Toxicol Appl Pharmacol. 2009 Sep 15;239(3):241-50. doi: 10.1016/j.taap.2009.05.028. Epub 2009 Jun 11.

PMID: 19523970

[A CaPful of mechanisms regulating the mitochondrial permeability transition.](#)

Di Lisa F, Bernardi P.

J Mol Cell Cardiol. 2009 Jun;46(6):775-80. doi: 10.1016/j.yjmcc.2009.03.006. Epub 2009 Mar 19.

PMID: 19303419 Review.

[See all similar articles](#)

Total 27 similar articles are listed here
on pubmed as well

Cell. 2008 Mar 21;132(6):1011-24. doi: 10.1016/j.cell.2008.01.028.

PMID: 18358813

Similar articles

Transport proteins (carriers) of mitochondria.

Wohlrab H.

IUBMB Life. 2009 Jan;61(1):40-6. doi: 10.1002/iub.139.

PMID: 18816452 Review.

Systematic analysis of the twin cx(9)c protein family.

Longen S, Bien M, Bihlmaier K, Kloeppel C, Kauff F, Hammermeister M, Westermann B, Herrmann JM, Riemer J.

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PMID: 19303419 Review.

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PubMed groups these using their text-neighboring algorithm (looking at shared MeSH terms and title/abstract keywords), which is why most of the 27 results focus on the SAM complex and outer mitochondrial membrane insertion

Q.7

An official website of the United States government [Here's how you know](#) ✓

NIH National Library of Medicine
National Center for Biotechnology Information

PubMed®

Nature[Journal] AND 2026[Date - Publication]

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RESULTS BY YEAR

2025 2026

PUBLICATION DATE

☒ 1 year
☐ 5 years
☐ 10 years
☐ Custom Range

TEXT AVAILABILITY

☐ Abstract
☐ Free full text
☐ Full text

ARTICLE ATTRIBUTE

2,937 results

Page 1 of 294

Filters applied: in the last 1 year. [Clear all](#)

☐ **Ketogenic diet mediates intestinal tumorigenesis through lipids not ketones.**
1 Shay JES, Chi F, Tzouanas CN, Han S, Zhang X, Ten Hoeve J, Williams KJ, Neptun S, Sever T, Fuentes I, Bhatia SN, Calibasi-Kocal G, Vander Heiden MG, Shalek AK, Yilmaz OH.
Cite **Nature.** 2026 Aug;656(8127):474-483. doi: 10.1038/s41586-026-10779-y. Epub 2026 Jul 15. PMID: 42457955

☐ **GPC3-specific dnTGFβRII-armoured CAR T cells for hepatocellular carcinoma.**
2 Zhang Q, Fu Q, Shen Y, Cao W, Jin G, Zhang Y, Wu J, Chen C, Wang H, Xu X, Sun K, Xue X, Bondanza A, Cao D, Chu N, Durham N, Cobbold M, Farsaci B, Giardino Torchia ML, Hao Y, Hong Y, Huang J, Humphries M, Jian Q, Moody G, Stone J, Sun L, Tu E, Wang F, Wang F, Yang Y, Yao Y, Zou A, Bai X, Liang T.
Cite **Nature.** 2026 Jul 15. doi: 10.1038/s41586-026-10786-z. Online ahead of print. PMID: 42457964

☐ **Diet-microbiome synergy underlies obesity-associated immunotherapy efficacy.**
3 Desharnais L, Swaby A, Messaoudene M, Doré S, Yu MW, Fiset B, Breton V, Ponce M, Hu Y, Wilson L, Sorin M, Wang Y, Dewar K, Pollak M, Elkrief A, Routy B, Walsh LA, Quail DF.
Cite **Nature.** 2026 Jul 8. doi: 10.1038/s41586-026-10750-x. Online ahead of print. PMID: 42420462

☐ **Mitochondrial metabolism and epigenetic crosstalk drive SASP.**
4 Martini H, Birch J, Marques EDM, Vittorelli S, Lagnado AB, Pirius N, Franco AC, Lee G, Han Y, Rowsey JJ

PubMed shows 2937 articles

Search Sources SciVal ? Create account

Advanced query ☐

EXACTSRCTITLE ("Nature") AND PUBYEAR = 2026

[Save search](#) [Set search alert](#) [Edit in advanced search](#)

Documents Preprints Secondary documents

22,924 documents found

Refine search

Search within results

Filters

Year

☒ Range ☐ Individual

from to

[Show less](#) [Full Text](#) [View at Publisher](#) [Related documents](#)

[Show abstract](#) [Full Text](#) [View at Publisher](#) [Related documents](#)

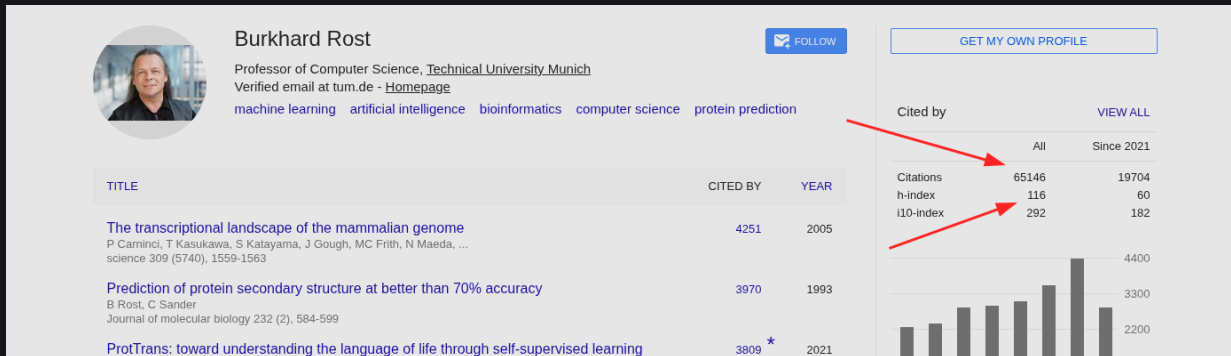
Document title	Authors	Source	Year	Citations
Minimalist terahertz wireless transceiver in integrated photonics	Guo Y, Zhang X, Zhang Y, ... Wang X, Shu H.	Nature Communications, 17(1), 4429	2026	0
CXCR3 is associated with T-cell-induced heart damage in acute rheumatic fever	Middleton F.M., McGregor R., Lorenz N., ... Carapetis J., Moreland N.J.	Nature Communications, 17(1), 4664	2026	0

SCOPUS shows 22,924 articles

[New comment](#)

the huge jump in Scopus makes sense since PubMed strictly indexes life sciences and biomedical journals, while Scopus indexes cross-disciplinary engineering, materials science, and physics literature as well

Q.8

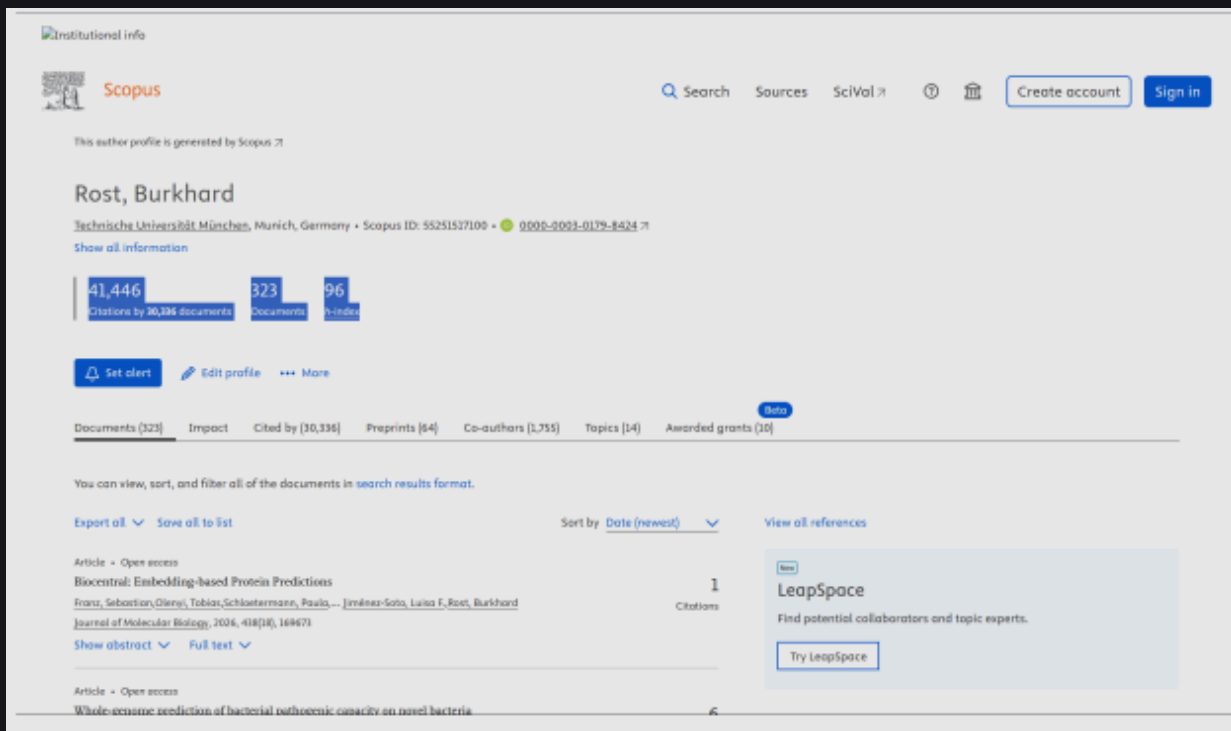


Scopus author profile for Burkhard Rost. The profile includes a photo, name, affiliation (Professor of Computer Science, Technical University Munich), and a list of keywords: machine learning, artificial intelligence, bioinformatics, computer science, protein prediction. A table lists his publications with citation counts and years. A bar chart shows the number of citations per year from 2010 to 2021.

TITLE	CITED BY	YEAR
The transcriptional landscape of the mammalian genome P Caminci, T Kasukawa, S Katayama, J Gough, MC Frith, N Maeda, ... science 309 (5740), 1559-1563	4251	2005
Prediction of protein secondary structure at better than 70% accuracy B Rost, C Sander Journal of molecular biology 232 (2), 584-599	3970	1993
ProtTrans: toward understanding the language of life through self-supervised learning	3809 *	2021

Cited by: All (65146), Since 2021 (19704).
h-index: 116, i10-index: 292.

on google scholar



Google Scholar profile for Burkhard Rost. The profile shows 41,446 citations by 30,334 documents and an h-index of 96. It lists his affiliation (Technische Universität München) and Scopus ID. A table shows his publications with citation counts and years. A bar chart shows the number of citations per year from 2010 to 2021.

Article	Citations
Biocentral: Embedding-based Protein Predictions Franz, Sebastian, Oliver, Tobias, Schlottermann, Paula, ... Jiménez-Goto, Luisa F., Rost, Burkhard Journal of Molecular Biology, 2016, 418(18), 169671	1
Whole-genome prediction of bacterial pathogenic capacity on novel bacteria	6

on pubmed

41,446 citations and 96 h-index

[New comment](#)

an h-index of 96 means at least 96 of the papers have been cited 96 or more times, which is a massive benchmark in computational structural biology

link to brenda

Information on EC 1.7.2.3 - trimethylamine-N-oxide reductase

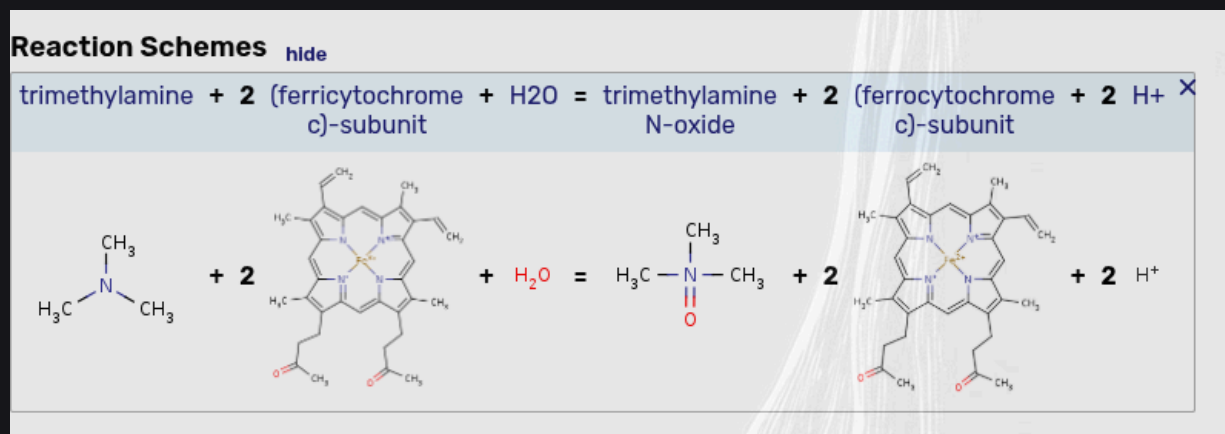
for references in articles please use BRENDA:EC1.7.2.3

EC Tree → 1 Oxidoreductases → 1.7 Acting on other nitrogenous compounds as donors → 1.7.2 With a cytochrome as acceptor → 1.7.2.3 trimethylamine-N-oxide reductase

this shows the enzyme class hierarchy

- ◆ Accepted Name: Trimethylamine-N-oxide reductase (cytochrome c), also commonly known as TMAO reductase.
- ◆ Enzyme Class Hierarchy:
 - ◆ EC 1 (Class): Oxidoreductases (enzymes that catalyze oxidation-reduction reactions).
 - ◆ EC 1.7 (Subclass): Acting on other nitrogenous compounds as donors.
 - ◆ EC 1.7.2 (Sub-subclass): With a cytochrome as an acceptor.
- ◆ Function / Catalytic Activity: This microbial enzyme catalyzes the chemical reduction of trimethylamine N-oxide (TMAO) into trimethylamine (TMA) as part of the electron transport chain.

reaction



[New comment](#)

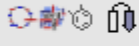
bacteria like *E. coli* use this as an alternative terminal electron acceptor under anaerobic respiration when oxygen isn't

around

Q.10

finding the EC number on brenda



 **6.3.5.4** asparagine synthase (glutamine-hydrolysing)

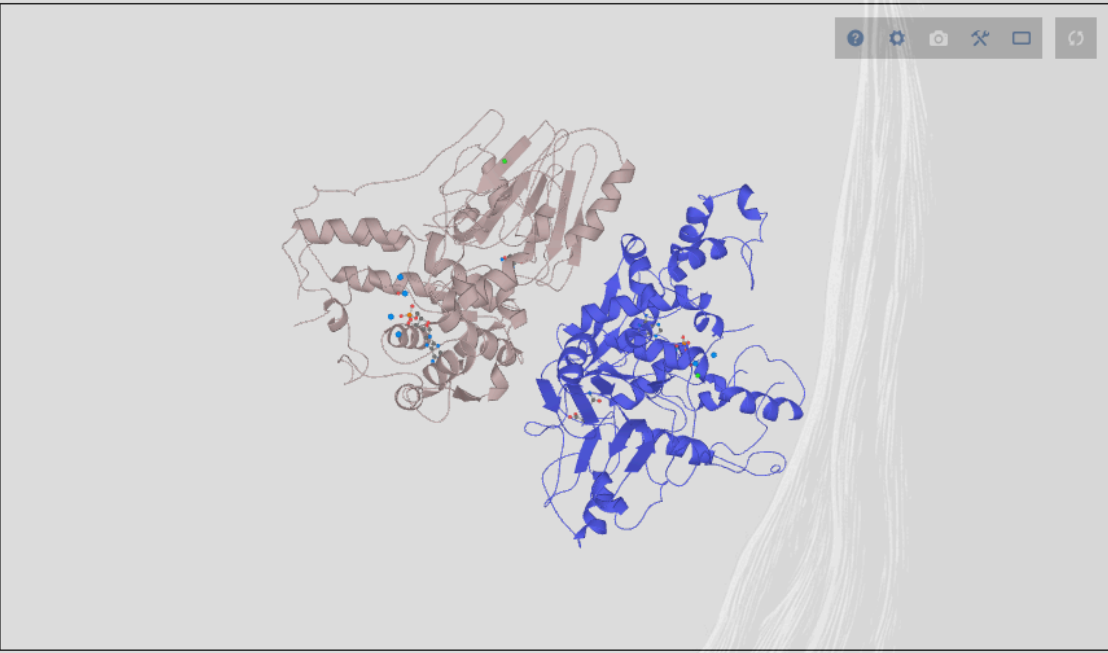
now searching it on ebi(thornton) gives us the enzyme

<https://www.ebi.ac.uk/thornton-srv/m-csa/entry/302/>

PDB
1ct9Q - CRYSTAL STRUCTURE OF ASPARAGINE SYNTHETASE B FROM ESCHERICHIA COLI (2.0 Å)  

 **Catalytic CATH Domains**
3.60.20.10  3.40.50.620  (see all for 1ct9Q )

Cofactors
Magnesium(2+) (1) Metal MACiE 



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and below it shows the catalytic residues and their roles

Catalytic Residues Roles

UniProt	PDB* (1ct9)		
Cys2 (N-term)	Ala1A (N-term)	Acts as a general acid/base to activate the cysteine nucleophile.	proton acceptor, proton donor
Leu51 (main-C)	Leu50A (main-C)	Helps stabilise the reactive intermediates formed.	hydrogen bond acceptor, electrostatic stabiliser
Thr322, Arg325	Thr321A, Arg324A	Bind and stabilise the phosphate groups of the ATP and reactive intermediates formed.	hydrogen bond donor, electrostatic stabiliser
Cys2	Ala1A	Acts as a catalytic nucleophile in the glutaminase domain reaction.	covalently attached, hydrogen bond acceptor, nucleofuge, nucleophile, proton acceptor, proton donor
Gly76 (main-N), Asn75	Gly75A (main-N), Asn74A	Forms the oxyanion hole that stabilises the reactive intermediates and transition states formed.	hydrogen bond donor, electrostatic stabiliser

so answer = Cys 1, Cys 2, Gly76, Leu 51, Thr 322, and Arg 325.

New comment

the Cys residues are primarily doing the heavy lifting for coordinating the molybdenum/iron-sulfur active center, while the others stabilize the transition state and hydrogen-bond with the TMAO substrate

Q.11

Search for
as
complete name

Homo sapiens

Taxonomy ID: 9606 (for references in articles please use ncbitaxon:9606)

current name

Homo sapiens Linnaeus, 1758

Genbank common name: **human**

NCBI BLAST name: **primates**

Rank: **species**

Genetic code: [Translation table 1 \(Standard\)](#)

Mitochondrial genetic code: [Translation table 2 \(Vertebrate Mitochondrial\)](#)

Lineage(full)

[cellular organisms](#); [Eukaryota](#); [Opisthokonta](#); [Metazoa](#); [Eumetazoa](#); [Bilateria](#); [Deuterostomia](#); [Chordata](#); [Craniata](#); [Vertebrata](#); [Gnathostomata](#); [Teleostomi](#); [Euteleostomi](#); [Scombroidei](#)

Felis catus

Taxonomy ID: 9685 (for references in articles please use ncbitaxon:9685)

current name

Felis catus Linnaeus, 1758

includes: *Korat cats* L.

homotypic synonym:

Felis silvestris catus

Genbank common name: domestic cat

just listing the data below now:

Organism	Scientific Name	Taxonomy ID	Number of Chromosomes (2n)
Human	<i>Homo sapiens</i>	9606	46
Cat	<i>Felis catus</i>	9685	38
Dog	<i>Canis lupus familiaris</i>	9615	78
Domestic guinea pig	<i>Cavia porcellus</i>	10141	64
Thale cress	<i>Arabidopsis thaliana</i>	3702	10

 New comment

Arabidopsis having only 2n = 10 (5 chromosome pairs) is one of the main reasons it became the universal model organism for plant genetics, since the genome is super compact

Q.12

The Entrez Programming Utilities (E-utilities) are a suite of server-side programs that serve as the public API for the NCBI Entrez system. They allow users to programmatically search, link, and retrieve data from across all NCBI Entrez databases (including PubMed, Nucleotide, Protein, and Gene) without needing to use the web browser interface.

example syntax:

```
esearch.fcgi?db=<database>&term=<query>
```

example:

- ◆ <https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi?db=pubmed&term=science> 
[journal]+AND+breast+cancer+AND+2008[pdat]

to fetch a fasta format:

<https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=nuccore&id=34577062,24475906&rettype=fasta&retmode=text> 

syntax is:

- ◆ db=nuccore: Specifies the database (e.g., nuccore for nucleotide, or protein).
- ◆ id=AY330867: The unique identifier or accession number of the record.
- ◆ rettype=fasta: Tells the server you want the specific FASTA layout.
- ◆ retmode=text: Tells the server to return plain text rather than XML.

New comment

E-utilities has a default rate limit of 3 requests/sec without an NCBI API key (or 10 requests/sec with a key), which is why programmatic pipelines usually cache UIDs using [esearch](#) first before batch downloading them with [efetch](#)

Q.13

a. Protein properties

NAR Database Summary Paper Category List

[Nucleotide Sequence Databases](#)
[RNA sequence databases](#)
[Protein sequence databases](#)
[Structure Databases](#)
[Genomics Databases \(non-vertebrate\)](#)
[Metabolic and Signaling Pathways](#)
[Human and other Vertebrate Genomes](#)
[Human Genes and Diseases](#)
[Microarray Data and other Gene Expression Databases](#)
[Proteomics Resources](#)
[Other Molecular Biology Databases](#)
[Organelle databases](#)
[Plant databases](#)
[Immunological databases](#)
[Cell biology](#)

[Compilation Paper](#)
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[Compilation Paper](#)
[Category List](#)
[Alphabetical List](#)
[Category/Paper List](#)
[Search Summary Papers](#)

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General sequence databases

Protein properties

[AAindex](#)
[BindingDB](#)
[Cybase](#)
[dbPTM](#)
[eProS](#)
[HHMD](#)
[iProLINK](#)
[MALISAM](#)
[MegaMotifbase](#)
[MemMoRF](#)
[Minimotif miner](#)
[MobiDB](#)
[MP:PD](#)
[PFD - Protein Folding Database](#)
[PIDD](#)
[PINT](#)
[PPD](#)
[PPT-DB](#)
[Proteome-pI 2.0](#)
[ProTherm](#)
[PRTAD](#)
[REFOLD](#)
[SwissSidechain](#)
[TOPPR](#)

1. UniProtKB (Universal Protein Resource): The central hub for the collection of functional information on proteins, with accurate, consistent, and rich annotation of protein properties.
2. Pfam (or InterPro): A large collection of protein families, represented by multiple sequence alignments and hidden Markov models (HMMs), used to determine protein domain properties.

b. Small molecules (Structure related)

Small molecules

BitterDB
ChEBI - Chemical Entities of Biological Interest
ChemBank
ChemDB
Crystallography Open Database
CSD - Cambridge Structural Database
EDULISS (currently unavailable)
FragmentStore
Hemolytik
Het-PDB Navi
HIC-Up
MMsINC
mVOC
PDB-Ligand
PubChem
R.E.DD.B.
SCRIPDB
SuperDrug
SuperScent
SuperToxic

1. PubChem: A massive database maintained by NCBI that provides information on the chemical structures, physical properties, and biological activities of small molecules
2. ChEBI (Chemical Entities of Biological Interest): A freely available dictionary of molecular entities focused on strictly "small" chemical compounds and their structural properties.

c. Cancer gene databases

Cancer gene databases

ArrayMap
BCCTBbp
BreCAN-DB
Cancer RNA-Seq Nexus
Cancer3D
CancerGenes
CancerPPD
Candidate Cancer Gene Database
CanGEM
CanSAR
CaSNP
cBioPortal
CCDB
ccmGDB
CellLineNavigator
ChimerDB
CIVICdb
ClinVar
CMPD
Colorectal Cancer Atlas
COLT-Cancer
COSMIC
CTDatabase
dbDEMC
dbDEPC
DIVAS
DriverDBv2
FusionCancer
HLungDB
HPTAA
IARC TP53 Database
IGDB.NSCLC
intoGen
ITTACA
Inc2cancer
MethHC
MethyCancer
MoKCa
MutationAligner
Network of Cancer Genes
OncoDB.HCC
OpenTein
Pancreas Expression
PRECOG and related resources
rSNPBase 3.0

1. COSMIC (Catalogue of Somatic Mutations in Cancer): The world's largest and most comprehensive resource for exploring the impact of somatic mutations in human cancer.
2. cBioPortal for Cancer Genomics: An open-access resource for interactive exploration of multidimensional cancer genomics data sets.

New comment

COSMIC focuses on curated somatic point mutations and gene fusions per sample, whereas cBioPortal is great for

Class Assignment

Question 1

Sequences:

- ◆ Sequence 1: ACAGACGTCCG (11 bases)
- ◆ Sequence 2: ACCGTCCG (8 bases)

Scoring Parameters:

- ◆ Match: +2
- ◆ Mismatch: -1
- ◆ Gap: -2

Recurrence Formulas:

- ◆ Diagonal: $F(i-1, j-1) + (+2 \text{ if match, } -1 \text{ if mismatch})$
- ◆ Up (Gap in Sequence 2): $F(i-1, j) - 2$
- ◆ Left (Gap in Sequence 1): $F(i, j-1) - 2$
- ◆ Cell Score: $F(i, j) = \max(\text{Diagonal}, \text{Up}, \text{Left})$

Generated by www.bioinformatics.org/

_	**_**	**A**	**C**	**C**	**G**	**T**	**C**	**C**	**G**
_	**0**	-2	-4	-6	-8	-10	-12	-14	-16
A	-2	**2**	0	-2	-4	-6	-8	-10	-12
C	-4	0	**4**	2	0	-2	-4	-6	-8
A	-6	-2	**2**	3	1	-1	-3	-5	-7
G	-8	-4	**0**	1	5	3	1	-1	-3
A	-10	-6	**2**	-1	3	4	2	0	-2
C	-12	-8	-4	**0**	1	2	6	4	2
G	-14	-10	-6	-2	**2**	0	4	5	6
T	-16	-12	-8	-4	0	**4**	2	3	4
C	-18	-14	-10	-6	-2	2	**6**	4	2
C	-20	-16	-12	-8	-4	0	4	**8**	6
G	-22	-18	-14	-10	-6	-2	2	6	**10**

Traceback:

Start from the bottom-right cell (10) and trace backward through the cells that yielded each optimal score until reaching the origin (0, 0).

optimal global alignment:

Plaintext

```
A C A G A C G T C C G
A C - - - C G T C C G
```

[New comment](#)

the 3 gaps in Sequence 2 (-6 penalty) are necessary because matching the entire tail **CGTCCG** yields +12, which heavily outweighs introducing a gap penalty across **AGA**

Question 2

The sequences are:

- ◆ Sequence 1: ACGTATCGCCGTATA (15 bases)
- ◆ Sequence 2: GATGCTCTCCGAAA (14 bases)

Scoring parameters:

- ◆ Match: +2
- ◆ Mismatch: -1
- ◆ Gap: -2

For the Smith-Waterman formula,

0 is included as a lower bound because this is a local alignment. If continuing the alignment produces a negative score, the score resets to 0 and a new alignment can begin at that position:

$$H(i, j) = \max(0, H(i-1, j-1) + \text{score}, H(i-1, j) - 2, H(i, j-1) - 2)$$

** *	** *	**G**	**A**	**T**	**G**	**C**	**T**	**C**	**T**	**C**	**C**	**G**	**A**	**A**	**A**
-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A	0	0	2	0	0	0	0	0	0	0	0	0	2	2	2
C	0	0	0	1	0	2	0	2	0	2	2	0	0	1	1
G	0	2	0	0	3	1	1	0	1	0	1	4	2	0	0
T	0	0	1	2	1	2	3	1	2	0	0	2	3	1	0
A	0	0	2	0	1	0	1	2	0	1	0	0	4	5	3
T	0	0	0	4	2	0	2	0	4	2	0	0	2	3	4
C	0	0	0	2	3	4	2	4	2	6	4	2	0	1	2
G	0	2	0	0	4	2	3	2	3	4	5	6	4	2	0
C	0	0	1	0	2	6	4	5	3	5	6	4	5	3	1
C	0	0	0	0	0	4	5	6	4	5	7	5	3	4	2
G	0	2	0	0	2	2	3	4	5	3	5	9	7	5	3
T	0	0	1	2	0	1	4	2	6	4	3	7	8	6	4
A	0	0	2	0	1	0	2	3	4	5	3	5	9	10	8
T	0	0	0	4	2	0	2	1	5	3	4	3	7	8	9
A	0	0	2	2	3	1	0	1	3	4	2	3	5	9	10

For Needleman–Wunsch, the traceback starts at the bottom-right corner. For Smith–Waterman, traceback starts at the highest score anywhere in the table (10) and stops when a cell with a score of 0 is reached.

optimal local alignment:

Plaintext

TCGCCGTATA

TCTCCG-AAA

 [New comment](#)

here the alignment naturally cuts off the noisy N-terminal overhangs (**ACGTA** in Seq 1 and **GATG** in Seq 2) since local alignment discards regions that drag the cumulative score down